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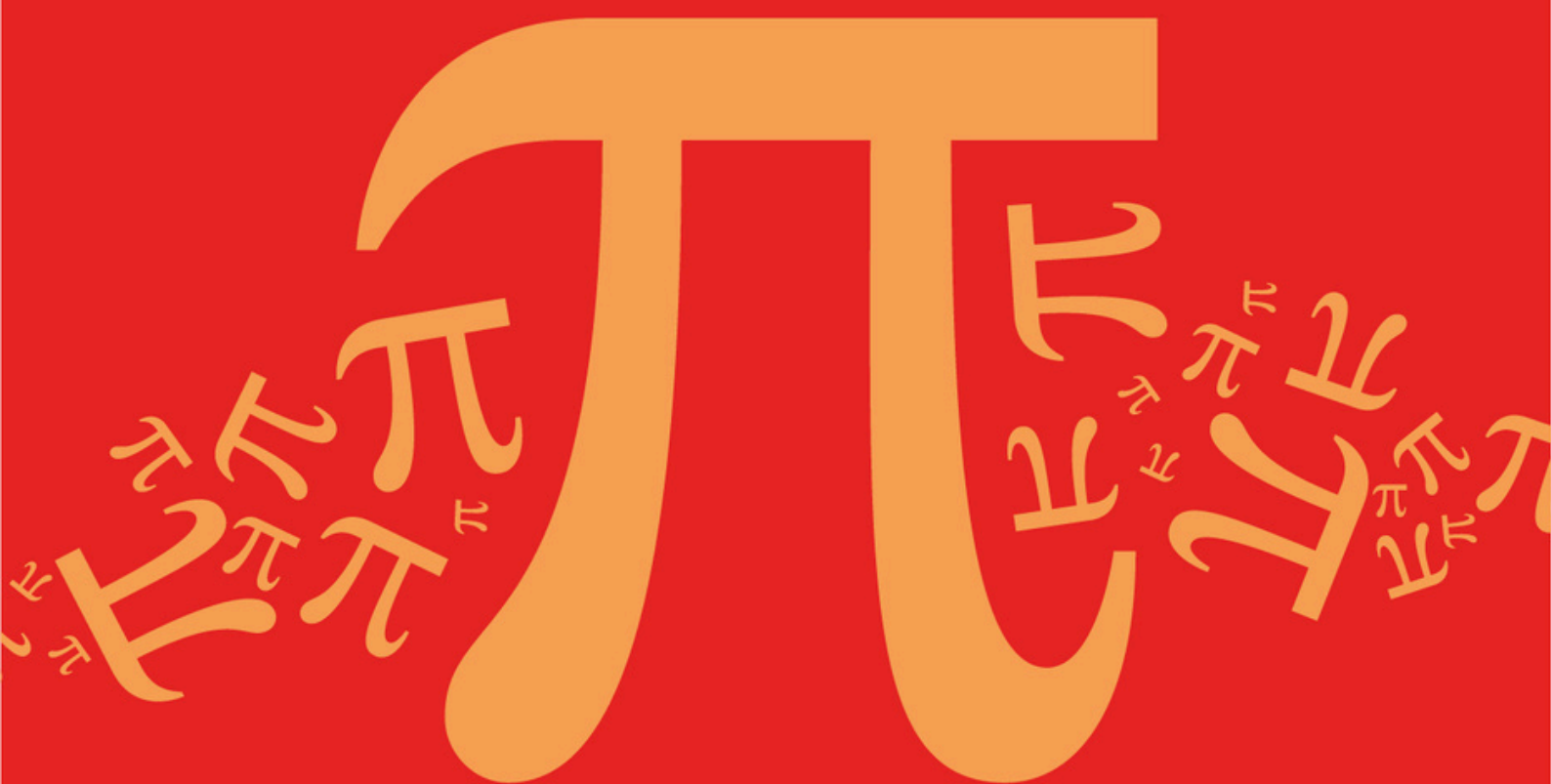
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Detailed mark scheme

Suitable for all boards

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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D3: Sums of Integers, Squares, and Cubes

Mark Scheme

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Topic	D: Further Algebra and Functions
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Demonstrates the rule is correct for $n = 1$	AO1.1b	B1	$\sum_{r=1}^1 r^3 = 1^3 = 1$ and $\frac{1}{4} \times 1^2 \times 2^2 = 1$ $\therefore$ it is true for $n = 1$
	States the rule is true for $n = k$ and adds $(k+1)^3$ to $\frac{1}{4}k^2(k+1)^2$	AO2.4	M1	Assume it is true for $n = k$ $\sum_{r=1}^k r^3 = \frac{1}{4}k^2(k+1)^2$
	Obtains $\frac{1}{4}(k+1)^2(k+2)^2$ from $\frac{1}{4}k^2(k+1)^2 + (k+1)^3$	AO2.2a	A1	$\sum_{r=1}^k r^3 + (k+1)^3 = \frac{1}{4}k^2(k+1)^2 + (k+1)^3$
	Completes a rigorous argument and explains how their argument proves the required result. This mark is only available if all previous marks have been awarded.	AO2.1	R1	$\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k^2 + 4(k+1))$ $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k^2 + 4k + 4)$ $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k+2)^2$ $\therefore$ it is also true for $n = k + 1$ True for $n = 1$, and true for $n = k \Rightarrow$ true for $n = k + 1$, then by induction it is true for all integers $n \geq 1$ AG

	Marking Instructions	AO	Marks	Typical Solution
(b)	Expresses LHS as summations of r^3 and r Ignore limits of the sums in this part only.	AO1.1b	B1	$\sum_{r=1}^{2n} r(r-1)(r+1) = \sum_{r=1}^{2n} (r^3 - r)$
	Expresses LHS in terms of n , using part (a) and $\sum r = \frac{1}{2}n(n+1)$	AO1.1a	M1	$= \sum_{r=1}^{2n} r^3 - \sum_{r=1}^{2n} r$
	Takes out $n(2n+1)$ as a factor or obtains $4n^4 + 4n^3 - n^2 - n$ Allow one slip in second bracket or one incorrect term in the expansion.	AO1.1a	M1	$= \frac{1}{4}(2n)^2(2n+1)^2 - \frac{1}{2}2n(2n+1)$ $= n^2(2n+1)^2 - n(2n+1)$
	Completes fully correct proof to reach the required result. This mark is only available if all previous marks have been awarded.	AO2.1	R1	$= n(2n+1)(n(2n+1) - 1)$ $= n(2n+1)(2n^2 + n - 1)$ $= n(2n+1)(2n-1)(n+1)$

Note: $n(n+1)(2n-1)(2n+1) = 4n^4 + 4n^3 - n^2 - n$			$= n(n+1)(2n-1)(2n+1)$ AG
Total 8 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
(a) Splits up the sum into separate sums $\sum ar^2 + \sum br + (\sum c)$ PI	AO3.1a	M1	$\sum_{r=1}^n (r+1)(r+2) =$ $\sum_{r=1}^n (r^2 + 3r + 2)$
Substitutes for the two sums $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ Allow one slip	AO1.1a	M1	$= \sum_{r=1}^n r^2 + \sum_{r=1}^n 3r + \sum_{r=1}^n 2$
States or uses $\sum_{r=1}^n 1 = n$ PI	AO1.2	B1	$S = \frac{n}{6}(n+1)(2n+1)$ $+ 3 \frac{n}{2}(n+1) + \sum_{r=1}^n 2$
Factorises out $(n+1)$ Allow one slip	AO1.1a	M1	$= \frac{n}{6}(n+1)(2n+1)$ $+ 3 \frac{n}{2}(n+1) + 2n$
Simplifies $(n+1)\{\frac{n}{2}(2n+1) + \frac{9n}{2} + 6\}$ to find second linear factor from 'their' quadratic FT 'their' quadratic provided all M1 marks have been awarded Allow one slip	AO1.1a	M1	$6 + 3 \sum_{r=1}^n (r+1)(r+2)$ Now $= 6 + \frac{n}{2}(n+1)(2n+1)$ $+ 9 \frac{n}{2}(n+1) + 6n$
Completes a rigorous argument to show the required result To obtain this mark factorising must be clearly seen and all previous marks obtained	AO2.1	R1	$= \frac{n}{2}(n+1)(2n+1) + \frac{9n}{2}(n+1) + 6(n+1)$ $=$ $(n+1)\{\frac{n}{2}(2n+1) + \frac{9n}{2} + 6\}$

			$= (n+1)(n^2 + 5n + 6)$ $= (n+1)(n+2)(n+3)$
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(b)	Chooses a multiple of 4 for n and obtains a correct numerical value/expression	AO2.4	E1	When $n = 4$, $6 + 3 \sum_{r=1}^n (r+1)(r+2) = (5)(6)(7)$
	Clear argument with concluding statement	AO2.3	E1	= 210 which is not a multiple of 12 so Alex's statement is false.
				Total 8 marks

Q3.

(a)
$$\sum_{r=1}^n 2r(2r^2 - 3r - 1) = \sum_{r=1}^n 4r^3 - \sum_{r=1}^n 6r^2 - \sum_{r=1}^n 2r = 4 \sum_{r=1}^n r^3 - 6 \sum_{r=1}^n r^2 - 2 \sum_{r=1}^n r$$

*Splitting up the sum into three separate sums.
PI by m1 line below.*

M1

$$= 4 \times \frac{1}{4} n^2 (n+1)^2 - 6 \times \frac{1}{6} n(n+1)(2n+1) - 2 \times \frac{1}{2} n(n+1)$$

Substitution of the three summations from FB into

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m1

$$= n^2(n+1)^2 - n(n+1)(2n+1) - n(n+1)$$

PI by later expressions

A1

$$= n(n+1)[n(n+1) - (2n+1) - 1]$$

Taking out factor $n(n+1)$ from correct expressions

m1

$$= n(n+1)[n^2 - n - 2]$$

A1

$$= n(n+1)(n+1)(n-2) (= n(n-2)(n+1)^2 \text{ (} p = -2, q = 1 \text{)})$$

A1

$$(b) \sum_{r=1}^{20} 2r(2r^2 - 3r - 1) = \sum_{r=1}^{20} 2r(2r^2 - 3r - 1) - \sum_{r=1}^{10} 2r(2r^2 - 3r - 1)$$

$$\sum_{r=1}^{20} \dots - \sum_{r=1}^{10} \dots$$

PI by next line (ft c's p & q)

M1

$$= 20(20 + p)(20 + q)^2 - 10(10 + p)(10 + q)^2$$

$$= 20 \times 18 \times 21^2 - 10 \times 8 \times 11^2 = 158760 - 9680 = 149080$$

NMS 0/2

A0 if not showing use of fully factorised form.

A1

2

[8]

Q4.

(a) $f(x) = 4x^3 - x - 540\,000$

$f(51) = -9447 (<0); f(52) = 22380 (>0);$

f(51) and f(52) both considered

M1

Since sign change (and f continuous), $51 < \alpha < 52$

All values and working correct plus relevant concluding statement involving '51' and '52'.

A1

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2

(b) (i) $S_n = \sum_{r=1}^n (2r-1)^2 = \sum_{r=1}^n 4r^2 - \sum_{r=1}^n 4r + \sum_{r=1}^n 1$

Splitting up the sum into separate sums. PI by m1 line below or better

M1

$$= 4 \frac{n}{6} (n+1)(2n+1) - 4 \frac{n}{2} (n+1) + \sum_{r=1}^n 1$$

Substitution of correct formulae from FB for the two summations

m1

$$= 4 \frac{n}{6} (n+1)(2n+1) - 4 \frac{n}{2} (n+1) + n$$

B1 for $\sum_{r=1}^n 1 = N$ stated or used

B1

A1

$$= \frac{n}{3} [2(2n^2 + 3n + 1) - 6(n + 1) + 3] = \frac{n}{3} [4n^2 - 1]$$

CSO

A1

5

(ii) $(6S_n = 2n[4n^2 - 1]) = 2n(2n - 1)(2n + 1)$

Terms in any order

B1

$(2n - 1)$, $2n$ and $(2n + 1)$ are consecutive integers

Terms must be identified and statement 'consecutive integers'

E1

2

- (c) $S_n = 1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2$ is sum of squares of first n odd numbers so need least N such that $S_N > 180\,000$

$$S_{52} = \frac{52}{3} [4 \times 52^2 - 1] = 187460 \text{ and } S_{51} = 176851$$

Either $\frac{n}{3} [4n^2 - 1] = 180000$ or $2N(2N - 1)(2N + 1) = 1080000$

*or S_{52} and S_{51} both attempted
(or = replaced by > or by $\geq$)*

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Smallest value of N is 52

CSO Fully and correctly justified.

NMS $N = 52$ scores 0 / 2

A1

2

[11]

Q5.

(a) $\sum r^2(4r - 3) = 4\sum r^3 - 3\sum r^2 \dots$

Splitting up the sum into two separate sums. PI by next line.

M1

$$= 4\left(\frac{1}{4}\right)n^2(n + 1)^2 - 3\left(\frac{1}{6}\right)n(n + 1)(2n + 1)$$

Substitution of the two summations from FB

M1

$$= n(n+1) \left[n(n+1) - \frac{1}{2}(2n+1) \right]$$

Taking out common factors n and $n+1$.

M1

Remaining expression eg our [...] in ACF not just simplified to AG

A1

$$\text{Sum} = \frac{1}{2}n(n+1)(2n^2-1)$$

Be convinced as form of answer is given, penalise any jumps or backward steps

A1

5

$$(b) \sum_{r=20}^{40} r^2(4r-3) - \sum_{r=1}^{19} r^2(4r-3)$$

Attempt to take $S(19)$ from $S(40)$ using part (a)

M1

$$= 20(41)(3199) - 9.5(20)(721) = 2623180 - 136990$$

$$\sum_{r=20}^{40} r^2(4r-3) = 2486190$$

2486190 ; Since 'Hence' NMS 0 / 2.

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SC $\sum_{r=1}^{40} r^2(4r-3) - \sum_{r=1}^{20} r^2(4r-3)$ clearly attempted and evaluated to 2455390 scores B1

A1

2

[7]

Q6.

$$(a) \sum r(3r+1) = 3\sum r^2 + \sum r$$

M1

$$= 3 \left(\frac{1}{6}n(n+1)(2n+1) \right) + \frac{1}{2}n(n+1)$$

correct formulae substituted

m1

$$= \frac{1}{2}n(n+1)(2n+1+1)$$

$m1$ for each factor (n and $n + 1$)

m1m1

$\dots = n(n + 1)^2$ convincingly shown
AG

A1

5

- (b) Correct expansion of $n(n + 1)^2$
and conclusion drawn (AG)

B1

1

[6]

Q7.

- (a) $\sum r^3 + \sum r = \frac{1}{4}n^2(n+1)^2 + \frac{1}{2}n(n+1)$
at least one term correct

M1

Factor n clearly shown
or $n + 1$ clearly shown to be a factor

m1

$$\dots = \frac{1}{4}n(n+1)(n^2 + n + 2)$$

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OE; A1 for $\frac{1}{4}$, A1 for quadratic

A1A1

4

- (b) Valid equation formed

M1

Factors n , $n + 1$ removed

m1

$$3n^2 - 29n - 10 = 0$$

OE

A1

Valid factorisation or solution
of the correct quadratic

m1

$n = 10$ is the only pos int solution

SC 1/2 for $n = 10$ after correct quad

A1

5

[9]

Q8.

(a) (i) $u_1 = S_1 = 1^2 \cdot 2 \cdot 3 = 6$
AG

B1

1

(ii) $u_2 = S_2 - S_1 = 42$
AG

B1

1

(iii) $u_n = S_n - S_{n-1}$

M1

$$= n^2(n+1)(n+2) - (n-1)^2n(n+1)$$

A1

$$= n(n+1)(4n-1)$$

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A1

3

(b) $\sum_{r=n+1}^{2n} u_r = S_{2n} - S_n$

M1

$$= (2n)^2(2n+1)(2n+2) - n^2(n+1)(n+2)$$

A1

$$= 3n^2(n+1)(5n+2)$$

AG

A1

3

[8]

Q9.

(a) $S_n = 3 \sum r^2 - 3 \sum r + \sum 1$

M1

Correct expressions substituted
At least for first two terms

m1

Correct expansions

A1

$\sum 1 = n$

B1

Answer convincingly obtained
 AG

A1

5

(b) $S_{2n} - S_n$ attempted
Condone $S_{2n} - S_{n+1}$ here

M1

Answer $7n^3$

A1

2

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[7]

Q10.

(a) Use of formula for $\sum r^3$ or $\sum r$

M1

n is a factor of the expression
clearly shown

m1

So is $(n + 1)$
ditto

m1

$S_n = \frac{1}{4} n(n+1)(n^2 + n - 12)$

A1

$$\dots = \frac{1}{4} n(n+1)(n+4)(n-3)$$

ft wrong value for k

A1F

5

- (b) $n = 1000$ substituted into expression
The factor 1004, or $1000 + 4$, seen

m1

Conclusion convincingly shown
not ' 2008×124749625 '

A1

Need $\frac{1000}{4}$ is even, hence conclusion
OE

2

[7]

Q11.

(a) (i) $(2r-1)^2 = 4r^2 - 4r + 1$

B1

1

(ii) $\sum (2r-1)^2 = 4\sum r^2 - 4\sum r + \sum 1$

M1

$$\dots = \frac{4}{3} n^3 - \frac{4}{3} n + \sum 1$$

m1A1

$$\sum 1 = n$$

B1

Result convincingly shown
AG

A1

5

- (b) Sum = $f(100) - f(50)$
M1 for 100 ± 1 and 50 ± 1

M1A1

$$\dots = 1\,166\,650$$

$$SC \ f(100) - f(51) = 1\,156\,449: 3/4$$

A2

4

[10]

Q12.

$$\Sigma(r^2 - r) = \Sigma r^2 - \Sigma r$$

M1

At least one linear factor found

m1

$$\Sigma(r^2 - r) = \frac{1}{6} n(n+1)(2n+1-3)$$

OE

m1

$$\dots = \frac{1}{3} n(n+1)(n-1)$$

A1

4

[4]

Q13.

$$(a) \quad \Sigma r^2(r-1) = \Sigma r^3 - \Sigma r^2$$

With attempt to use the given formulae

M1

Good progress with expansion

or use of common factors

m1

Factors n and $n + 1$ found

Allow verification here

A1

$$\dots = \frac{1}{12} n(n^2 - 1)(3n + 2)$$

Convincingly shown (AG)

A1

4

(b) Use of $f(11) - f(3)$ in above

M1 for $f(11) - f(4)$

M1

$f(11) = 3850$

PI by correct answer

A1

$f(3) = 22$ (so answer is 3828)

ditto

A1

3

[7]



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Examiner reports

Q1.

The vast majority of students clearly knew how to structure a proof by induction, but only a third of the students could produce a fully correct proof. Almost all attempts included a confirmation of the initial case of $n = 1$, and many of these then assumed the identity was true for $n = k$. Good students were able to successfully manipulate this equation into a form which satisfied the $n = k + 1$ case. Some of these, however, resorted to expanding their quartic expression and then demonstrating that it was equivalent to the required quartic. Although an acceptable method, this was a time consuming approach, and prone to error. However, those who succeeded in this usually went on to produce a satisfactory proof by induction, although a minority demonstrated a lack of understanding of the requirements of a proof by induction.

The majority of students were able to make some progress in part (b), correctly substituting $2n$ into the appropriate formulae, but some of these then struggled with the algebra. Again, a minority of students opted to expand their quartic, instead of factorising, but then errors often crept in.

Q3.

This question was another good source of marks. A large majority of students correctly multiplied out the bracket, split the sum into three relevant separate sums and applied the formulae from the formulae booklet correctly. Better students then took out the relevant common factors and factorised the resulting quadratic correctly to reach the correct answer. Those students who expanded the brackets to reach a quartic in n were slightly less successful, but to their credit many did manage to reach the correct answer albeit in a more time consuming manner. A common mistake amongst the weakest students, which resulted in no marks being scored for part (a), was to initially write the given left-hand side

as $\sum 2r \left(\sum 2r^2 - \sum 3r - \sum 1 \right)$ before using the formulae booklet.

In part (b) most students scored both marks, but some lost the accuracy mark because they did not show sufficient detail in their working to indicate that 'Hence' had been applied. A more serious error, which resulted in the loss of both marks, was to use

$$\sum_{r=1}^{20} \dots - \sum_{r=1}^{11} \dots$$

Q4.

This question proved to be the most challenging question on the paper. In part (a) most students scored the method mark but a significant number failed to score the remaining mark. Some statements had the root lying between -9447 and 22380 , others failed to specify '51' and '52' and others failed to mention the sign change each of which resulted in the loss of the second mark. In part (b)(i) many scored the method marks but a

significant proportion did not use $\sum_{i=1}^n 1 = n$, just using 1 instead of n . Full correct answers for (b)(i) were seen relatively frequently but that cannot be said for the final two demanding parts of the question. In part (b)(ii) many failed to correctly factorise $4n^2 - 1$ within their correct expression $2n(4n^2 - 1)$ and very few of those that did could correctly identify the three terms as $(2n - 1)$, $(2n)$ and $(2n + 1)$ and make the necessary statement that these were 'consecutive integers'. There was slightly more success in the

final part of the question but a common wrong approach involved starting with $\frac{n}{6}(n+1)(2n+1) > 180000$.

Q5.

There were some excellent solutions seen in answers to this question on series which used at an early stage the fact that $n(n+1)$ was a factor of the two expressions. However a significant number of candidates expanded out to reach a quartic expression and then jumped too quickly, without sufficient intermediate steps being shown, to the printed form of the answer. It was not uncommon to see incorrect quartics being followed by the 'correct' printed form of the answer. The most common error, which resulted in the loss of the final accuracy mark in part (a), was to write ' $= 2n(n+1)(2n^2-1)$ ' having correctly

reached ' $n(n+1)(n^2 - \frac{1}{2})$ '.

In part (b) the most common error was to subtract $S(20)$ rather than $S(19)$ from $S(40)$.

Q6.

In many cases candidates proved the result given in part (a), often by expanding fully before attempting to find the factorised form. Luckily in this case it was not very hard to obtain the factors from the expanded form, particularly as the answer was given. Some candidates, however, lost credit through not showing steps at the end, which they may have thought unnecessary as the answer was 'obvious', but this overlooked the importance of good examination technique when the answer is printed on the question paper. For a similar reason, some candidates lost the one mark available in part (b), no doubt thinking that the result was 'obvious' and not writing enough steps.

Q7.

Candidates who were accustomed to look for common factors — an approach almost always needed in questions on this topic — were able to obtain high marks in both parts, though some of these candidates surprisingly failed to solve the quadratic in part (b). Candidates who preferred to expand and simplify everything and then hope to spot some factors were often successful in part (a) but could not realistically hope for more than one mark in part (b).

Q8.

This question proved to be quite discriminating. Either candidates realised what they were asked to do and scored full marks, or the notation puzzled them and they were unable to proceed beyond part (a)(ii), thinking that the answer to this part should be 48 rather than 42. Some of the weaker candidates, whilst realising what to do, failed to take out common factors in their algebraic manipulation in parts (a)(iii) and (b) with the result that correct answers were written down after incorrect algebra.

Q9.

Part (a) of this question was very well answered. The algebra needed was not quite as heavy as is sometimes seen in questions on this topic, and the fact that the answer was given seemed to steer candidates smoothly to a correct solution. Only a small proportion of candidates wrote 1 for $\Sigma 1$. In part (b), only a minority of candidates showed any awareness of the need for a subtraction, but those who did show this awareness usually found the right answer with very little difficulty.

Q10.

- (a) As in the previous question, many candidates were not sufficiently familiar with the techniques needed to carry out the necessary manipulation efficiently. It was noticeable that many candidates still obtained full marks despite their failure to spot the quick method of taking out common factors at the earliest opportunity. A common mistake was to omit the numerical factor $\frac{1}{4}$, or to replace it with some other number such as 4.
- (b) Good attempts at this part of the question were few and far between. Many candidates made no attempt at all. Some came to a halt after replacing n by 1000 in their answer to part (a). Those who found a factor 1004 usually went on to try to explain how this would lead to a multiple of 2008, but more often than not their arguments lacked cogency.

Q11.

The simple request in part (a)(i) seemed to have the desired effect of setting the candidates along the right road in part (a)(ii). As usual many candidates struggled with the algebra but made reasonable progress. They could not hope to reach the printed answer legitimately if they equated $\sum 1$ to 1 rather than to n . Very few, even among the strongest candidates, achieved anything worthwhile in part (b), most using $n = 200$ instead of $n = 100$ at the top end. No credit was given for simply writing down the correct answer without any working, as the question required the candidates to use the formula previously established rather than summing the numbers directly on a calculator.

Q12.

Fortunately for those candidates who failed to extract the common factors, the algebra was not too heavy for them to expand both of the well-known expressions and to arrive at the correct answer without too much trouble. A mistake which occurred distressingly often was to 'drop' a numerical factor such as $1/6$, as if the candidates were solving an equation.

Q13.

Most candidates realised that a subtraction of the two given expressions was required. The work in part (a) was disappointingly pedestrian, though the examiners had to admire the tenacity of many candidates who failed to extract the common factors from the two terms, yet persevered with some very heavy algebra to reach the conclusion legitimately. Candidates should expect to be penalised severely for claiming to have reached the answer when in fact they have done nothing of the sort. The response to part (b) was mixed. Many candidates simply substituted $n = 11$, or sometimes $n = 7$, into the result from part (a). The most common mistake was to subtract the result with n equated to 4 instead of 3.